

Quadratic formula



1

a $a = 8, b = -3, c = 0$

b $a = 8, b = 0, c = 3$

c $a = 5, b = 6, c = -10$

d $a = 3, b = -17, c = 9$

2

a $x = -7, -4$

b $x = -\frac{1}{2}, -1$

c $x = 3, -\frac{5}{4}$

3 -2

4

a -9

b 0

5

a $x = \frac{5 \pm \sqrt{33}}{2}$

b $x = \frac{1 \pm \sqrt{161}}{8}$

c $x = \frac{-15 \pm \sqrt{193}}{4}$

6

a $x = 0.405, -7.405$

b $x = -0.260, -1.540$

7 $x = 0.4, -3.3$

8

a $x = \pm\sqrt{7}$

b $x = \pm 2.6$

9

a $x = 5, -1$

b $x = -4, -5$

Discriminant

10 That there is one real solution.

11 $b^2 - 4ac > 0$

- 12 a i 0 ii 1 b i -76 ii 0  i 1 ii 2

- 13 a Yes b No c Yes

- 14 a 0 b The roots are equal and rational.

- 15 a 256 b The roots are unequal and rational.
c $x = 12, -4$

Quadratics with complex solutions

16 0

17 0

- 18 a $x = -1 \pm 2i$ b $x = 3 \pm 4i$ c $x = -5 \pm \sqrt{3}i$ d $x = 2 \pm 5i$
e $x = 5 \pm \sqrt{2}i$ f $x = -\frac{5}{2} \pm \sqrt{2}i$ g $x = \frac{1}{2} \pm \sqrt{3}i$ h $x = \frac{2}{3} \pm \frac{2}{3}i$
i $x = -5 \pm \sqrt{15}i$ j $x = 2 \pm 4\sqrt{2}i$ k $x = -\frac{4}{3} \pm \sqrt{11}i$

19 $x = -\frac{7}{16} \pm \frac{\sqrt{47}}{16}i$

- 20 a Yes b No c No d Yes

- 21 a 0 b 0 c 2

22 $m > -1$

23 $n = \pm 9$



24 $k = 1$

25

a $64 - 8k$

b

i $k = 8$

ii $k \leq 8$

iii $k > 8$

iv $k < 8$

26

a $52 - 12k$

b

i $k = \frac{13}{3}$

ii $k \leq \frac{13}{3}$

iii $k > \frac{13}{3}$

iv $k < \frac{13}{3}$

Discriminant and parabolas

27 None

28

a $m < -\frac{27}{4}$

b $m = -7$

29 $k = 16$

30



a

i $a > 0$

ii $b^2 - 4ac > 0$

b

i $a < 0$

ii $b^2 - 4ac = 0$

c

i $a < 0$

ii $b^2 - 4ac < 0$

d

i $a > 0$

ii $b^2 - 4ac < 0$

e

i $a > 0$

ii $b^2 - 4ac < 0$

f

i $a > 0$

ii $b^2 - 4ac > 0$

g

i $a < 0$

ii $b^2 - 4ac = 0$

h

i $a < 0$

ii $b^2 - 4ac > 0$

i

i $a > 0$

ii $b^2 - 4ac = 0$

j

i $a < 0$

ii $b^2 - 4ac < 0$

k

i $a < 0$

ii $b^2 - 4ac = 0$

l

i $a < 0$

ii $b^2 - 4ac > 0$

m

i $a < 0$

ii $b^2 - 4ac < 0$

n

i $a > 0$

ii $b^2 - 4ac = 0$

o

i $a < 0$

ii $b^2 - 4ac > 0$

p

i $a > 0$

ii $b^2 - 4ac < 0$

Applications

31

a $0 < p < q$

c $8q(q - p)$

e $x = 3p$

b $x^2 + 2(p - q)x + p^2 - q^2 = 0$

d Two real solutions

32

a $200 = -5t^2 + 56t + 26$

b $x = \frac{28}{5} \pm \frac{\sqrt{86}}{5}i$

c The solution is not a real number, so the cannon-ball will not reach a height of 200 meters.

33 $x = 8.7$